DEPARTMENT OF AGRICULTURAL & BIOSYSTEMS ENGINEERING
COLLEGE OF ENGINEERING
KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

PERISHABLE CROP STORAGE



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STORAGE OF PERISHABLE CROPS

- Perishable crops are crops with moisture content above 60%. Examples of perishable crops are cassava, tomatoes, banana and oranges.
- Losses of horticultural crops due to improper storage and handling can range from 10 to 40%.
- Due to their high moisture content, perishables are expensive to store and very perishable under ambient conditions.

- The aim of storage is to provide an environment which slow down the rate at which decay processes occur.
- This is usually done by cooling the produce and by controlling the humidity and the composition of the atmosphere in the store.
- Cooling by refrigeration can substantially reduce the rate at which food will deteriorate.
- Low temperatures slow down the growth of microorganisms and the rate of chemical (including enzymic) changes in food.
- These are two of the main causes of food spoilage.

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Temperature and Storage

Refrigerated storage retards the following elements of deterioration in perishable crops:

- aging due to ripening, softening, and textural and color changes;
- undesirable metabolic changes and respiratory heat production;
- moisture loss and the wilting that results;
- spoilage due to invasion by bacteria, fungi, and yeasts;
- undesirable growth, such as sprouting of potatoes

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Recommended storage temperatures

1-4^{0}C (34-39^{0}F)	5-9^{0}C (41-48^{0}F)	>10^{0}C (58^{0}F)
Apple	Avocado (Temperate)	Avocado (Tropical)
Asparagus	Zucchini	Pumpkin
Broccoli	Egg Plant	Tomato
Grapes	Cucumber	Papaya
Lettuce	Peppers	Mango
Mushroom	Potato	Banana
Peach &Plum		Sweet Potato
Berry fruits		Pineapple

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The four most common cooling methods

Products may be cooled by means of:

1. Cold air
   a. Room cooling
   b. Forced air cooling
2. Cold water
   a. Contact icing
   b. Hydro cooling
3. Vacuum cooling
4. Evaporation cooling

Cold Air

Room cooling

- The most commonly used technique. The produce are in boxes, cartons, bulk containers, mobile containers and other packages exposed to cold air in a normal cool store.
- The product is cooled and stored at the same time.There is a refrigeration unit attached to the store.

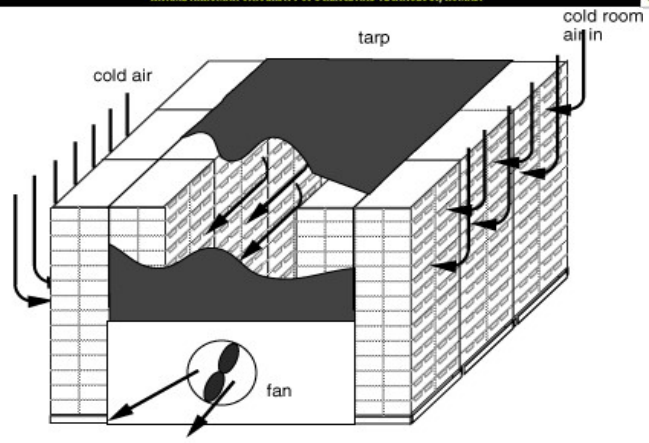
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Cold Air

Forced air cooling

- Cold air is force through the packages and around each piece of the product.
- Forced air cooling can cool the product in a shorter time than room air cooling since room cooling only removes heat from the surface of the product whiles forced air cooling removes heat from both the surface and within of the product.

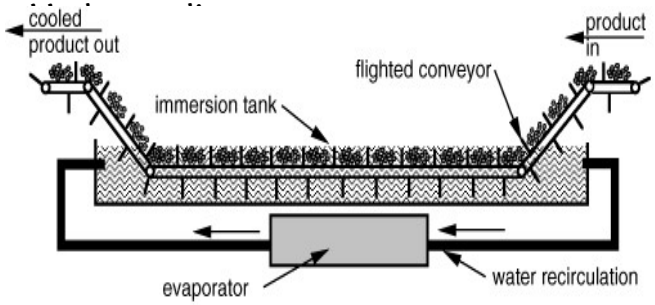
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Hydro Cooling

- In hydro cooling, water acts as heat transfer medium.
- The principle behind the use of this method is that the heat capacity of water is far greater than that of air.
- Produce are passed under cold showers in a moving conveyor. The product is cooled and clean.



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- **Contact Icing**

Ice is crushed and the product placed in it or the ice is placed on the product.

- **Vacuum Cooling**

It is a special technique used for vegetables.

It is uniform cooling at low pressure.

The product is loaded into a sealed container and the pressure is reduced considerably.

At this pressure, the produce is cooled by evaporation of water from the surface of the produce.

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Evaporative cooling

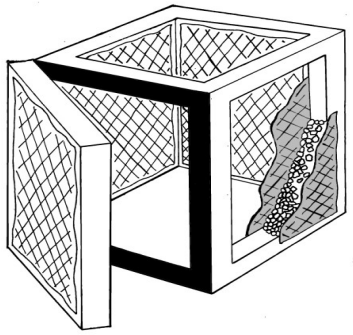
- It is a technique by which dry air is cooled by blowing it across a wet surface. It is effective in dry climates (areas where relative humidity is low).
- Evaporative cooling has low energy cost.

Two methods are used:

- Humidity cooled air for cooling.
- Misting – spraying water into air.

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Evaporative cooling box



CONTROL/MODIFIED (MA/CA) ATMOSPHERE STORAGE

- It involves modification of the storage environment by limiting oxygen and increasing the carbon dioxide concentration in the storage atmosphere.

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MA/CA Atmospheres

- Controlled or modified atmospheres (CA or MA) are terms implying the addition or removal of gases from storage rooms, or small-size food packages to manipulate the levels of oxygen, carbon dioxide, nitrogen, ethylene, etc., and achieve an atmospheric composition different from that of normal air around the food.

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MA/CA Atmospheres

- The term modified atmosphere is used when the composition of the storage atmosphere is not closely controlled (i.e. plastic packages), whereas controlled atmosphere generally indicates more precise monitoring and control of gaseous composition.

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Application of Controlled Atmosphere storage

The application of controlled-atmosphere packaging and storage is an attempt to

- retard physiological processes (such as respiration, ripening and deterioration) and
- minimize handling injuries and microbial infection to maintain optimum quality and extend the shelf life of fruits and vegetables.

Control conditions

- The oxygen content is reduced below 10% whiles the carbon dioxide concentration is increased to about 4 – 5%.
- A temperature of about 5ºC and oxygen concentration of about 2.5% reduces respiration by 50%. However, there should be sufficient oxygen to prevent anaerobic respiration.

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- For apples at 5^0C, O_2 levels of 2.5% halve the respiration rate.
- Increases of CO_2 levels also reduce respiration, but if these are too high then anaerobic effects can occur.
- Levels differ widely between crops, and some do not respond at all.

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COOLING CAPACITY CALCULATIONS

- Fruit must be cooled as quickly as is economically possible.
- In order to do this the capacity of the cooling plant needs to be calculated.
- This requires a knowledge of the internal energy which will need to be removed in the form of heat. (Internal energy is referred to as enthalpy)

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COOLING CAPACITY OF REFRIGERATION PLANTS

1. **Product load (Field heat)**

- Product load: it is the load energy which must be removed to reduce the temperature of the produce to the desired storage temperature.

- The product load, Hp is determined as:

- $Hp = \frac{2.8MS(T_f - T)}{t_s}$

Hp – watts (W)
t_s – Time(s)
M – Mass of produce/crop (kg)
S – Specific heat capacity of the produce J/kg^{0}C
T – Refrigeration temperature
T_f –Fruit /crop temperature

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QUESTION

Crop at a temperature of 25^{0}C is to be precooled to 0^{0}C in 12 hours. The heat capacity of the product is 3650J/kg^{0}C. The weight of the product is 15tons.

Determine the product load.

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Solution

- $Hp = \frac{2.8MS(T_f - T)}{t_s} = \frac{2.8 \times 15 \times 1000 \times 3650 \times (25-0)}{12 \times 3600}$

$= 88.7kW$

Hp – watts (W)
t_s – Time(s)
M – Mass of produce/crop (kg)
S – Specific heat capacity of the produce J/kg^{0}C
T – Refrigeration temperature
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RESPIRATION LOAD

- The crop is a living breathing product and gives off heat as it breathe. Respiration load is the quantity of heat given off during respiration.
- It depends on the metabolic activity and also temperature. The respiration load or respiration heat is given by:

$$Hr = MH$$

- M – Product mass (kg)
- H – Metabolic heat/ respiration heat/ heat given off (W/kg)

Question

The metabolic heat at O^0C for ripe tomatoes is 0.05, 0.009 for onions, 0.012 for apples, and 0.009 for oranges.

Given 10 tonnes of each product, determine which of the product will give the lowest respiration load.

WALL LOAD (Hw) - HEAT TRANSFER THROUGH THE WALLS OF THE COLD STORE

$$Hw = UA (Tx - T)$$

Tx = Temperature of medium- Outside temperature (^0C)

T = Inside temperature- desired storage temperature (^0C)

Hw = Wall load (W)

U = Heat transfer co-efficient ($W/m^2\ ^0C$)

A = Total outside surface area of the cold store (m^2)

Question

- Given the dimension of a Cold store to be 9m x 9m x 4m
- Calculate the total wall load for outside temperatures of 30ºC, under floor temperatures of 20ºC and a floor temperature of 0ºC.The heat transfer of the various mediums is as given
- Walls – 0.241 $W/m^2ºC$
- Ceiling – 0.21 $W/m^2ºC$
- Floor – 0.35 $W/m^2ºC$

Air change load:

Due to
- air itself
- water component

Air change load: Ha =Ma (ha -h) + MwSw (Ta -T)

Air change load

Where

M_a is the mass of air admitted in kg per second

h_a is the enthalpy of the outside air (J/kg),

h is the enthalpy of the air at the conditions inside the cooler,

M_w is the mass of water which condenses out (kg/s)

S_w is the specific heat of water.

- The enthalpy and the mass of water are determined from the psychometric chart. This is a chart which gives us the properties of moist air at atmospheric pressure, provided we know the relative humidity and the temperature.

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Other Loads

- We allow for other heat loads produced by fans, motors, people, and lights. These loads are also hard to estimate and a sensible guess is appropriate.
- Some fans will be needed to circulate the cooling through the crop, and so the heat load from these at least can be estimated.

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EXAMPLE

Calculate the size of a refrigeration plan required to cool 15 tonnes of apples to 0°C from 20°C in 12 hours, in a pre-cooler. Use the following data:

Respiration rate for apples at 20 °C = 0.092 W/kg, U for cooler walls and ceiling =0.241 $Wm^{-2}°C^{-1}$, outside temperature 29.5 °C, U for floor=0.35, Underfloor temperature 14 °C, size of cooler 9m x 9m by 4m high, humidity of out air = 70%, assume one air change per hour.

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Solution

First we will calculate the heat loads, and then find the minimum surface conductance coefficient required to extract the heat from the apples,

a. The product load is given by:

P_p = {2.08 x 15,000 x 3650 x (20 - 0)}/12 x 3600

= 52.73 kW

b. The respiration load when the apples are first put in is:

- P_r = 15,000 x 0.092 = 1.38 kW When they have cooled to 0°C the load be 15,000 x 0.012 = 0.18 kW.
- Thus the load will drop as cooling proceeds, and in storage the lower value can be used. However for the pre-cooler, the greatest value is appropriate i.e., 1.38 kW.

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c. For the ceiling, walls, and floor respectively,
The heat load is:

$P_{wceiling}$ = 0.241 x9x9x(29.5-0)= 576 W
P_{wwall} = 0.241 x 9 x 4 x (4) x (29.5 - 0) = *1024*
P_{wfl} = 0.35 x 9 x 9 x (14 - 0) = 397 W
The total wall gain is thus the sum of these or 1.997 kW.

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d. Calculation of the air change load requires the use of the psychrometric chart.

- First the volume of air inside the cooler can be found by finding the point representing the moist air at 0°C and 100% RH, and reading the value of the volume from the near vertical lines (i.e., lines parallel to AG in the inset).
- *At 0°C*, one kg of dry air occupies a volume of 0.778 m^3. Hence for an air change of 1, the mass of air is:

M= (9 x 9 x 4) /0.778= 416.5kg

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CHILLING INJURY

- Some fresh produce are susceptible to injury at low but non-freezing temperature. Such crops are mostly of tropical or sub-tropical origin. A few temperate crops may be affected. Symptoms of chilling may not be developed until the produce is removed from cold storage to normal ambient temperature.
- Effect of Chilling Injury:
  - Discolouration- Internal or external
  - Abnormal ripening- e.g. uneven ripening
  - Increase in decay- activity by micro-organisms
  - Sunken spots

CHILLING INJURY

- Both temperature and duration of exposure are involved in the development of chilling injury.
- Damage may occur in a short time if temperatures are considerably below the threshold level.
- Products may be able to withstand temperatures a few degrees in the critical zone for a longer time before injury becomes irreversible.

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FREEZING INJURY

- Freezing injury occurs when ice crystals form in the tissues. Cultivars, locations, and growing conditions may affect the freezing point.
- To be on the safe side, the highest temperature at which freezing of a specific commodity may occur should be used as a guide for recommending the optimum storage temperature.